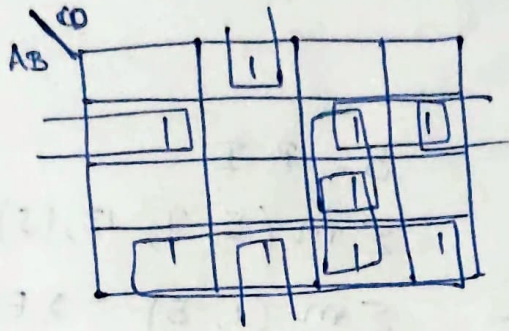


2) Determine Prime implicants for the function
 $\Sigma m(1, 4, 6, 7, 8, 9, 10, 11, 15)$



1 quad + 5 Pairs

$$\Rightarrow \Sigma m(8, 9, 10, 11)$$

$$\Rightarrow \Sigma m(10, 15), \Sigma m(5, 7)$$

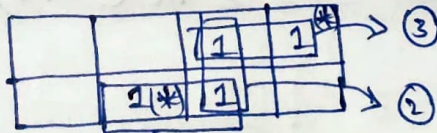
$$\Sigma m(7, 6)$$

$$\Sigma m(4, 6), \Sigma m(1, 1)$$

(*) Essential Prime implicants :- (★★★★)

The group which covers at least one min-term that can't be covered by other group / Prime Implicants.

Eg:- ①



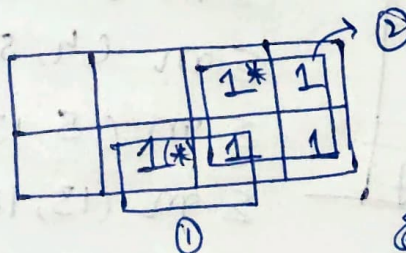
$$\textcircled{1} \rightarrow \Sigma m(5, 7), \Sigma m(13, 15)$$

$$\Sigma m(5, 7), \Sigma m(2, 3),$$

$$3 - \text{P.I. } \Sigma m(3, 7)$$

$$2 - \text{E.P.I.}$$

②



$$2 - \text{P.I. } \Sigma m(2, 3, 6, 7)$$

$$2 - \text{E.P.I. } \Sigma m(5, 7)$$

$$\cancel{\Sigma m(5, 7)} \quad \cancel{\Sigma m(3, 7)}$$

③

			1	1
1	1	1	1	

4 - P.I

$$\begin{aligned} \Sigma m(4,5) \\ \Sigma m(5,7) \\ \Sigma m(3,7) \\ \Sigma m(2,3) \end{aligned} \rightarrow \text{EPI}$$

④

		1	1	
	1	1	1	
1	1	1		
1				1

5 - P.I

$$\begin{aligned} \Sigma m(5,7,13,15) &\rightarrow \text{EPI} \\ \Sigma m(1,5) &\rightarrow \text{EPI} \\ \Sigma m(12,13) \\ \Sigma m(8,12) \\ \Sigma m(8,10) &\rightarrow \text{EPI} \end{aligned}$$

4 - P.I

$$\begin{aligned} \Sigma m(0,2,8,10) &\rightarrow \text{EPI} \\ \Sigma m(13,15) &\rightarrow \text{EPI} \\ \Sigma m(5,13) &\rightarrow \text{EPI} \\ \Sigma m(2,3) &\rightarrow \text{EPI} \end{aligned}$$

9 - P.I

$$\begin{aligned} \Sigma m(0,2,8,10) &\rightarrow \text{EPI} \\ \Sigma m(4,5) \\ \Sigma m(5,13) \\ \Sigma m(13,15) \\ \Sigma m(10,11) \\ \Sigma m(2,3) \end{aligned}$$

$$\Sigma m(3,11)$$

$$\begin{aligned} \Sigma m(0,1) \\ \Sigma m(11,15) \end{aligned}$$

②

1	1			
	1	1		
			1	
				1

5 - P.I

$$\begin{aligned} \Sigma m(0,1) &\rightarrow \text{EPI} \\ \Sigma m(1,5) \\ \Sigma m(5,7) \\ \Sigma m(7,15) \\ \Sigma m(15,11) &\rightarrow \text{EPI} \end{aligned}$$

Redundant Prime Implicants :- (RPI)

Prime implicants whose each of the min-term is a part of some essential Prime Implicant.

Selective Prime Implicants :- (SPI)

Neither EPI, Nor RPI.

Ex:-

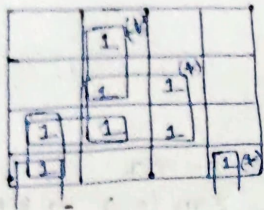
①

1	1		
	1	1	

3 - P.I

$$\begin{aligned} \Sigma m(0,1) &\rightarrow \text{EPI} \\ \Sigma m(1,5) &\rightarrow \text{RPI} \\ \Sigma m(5,7) &\rightarrow \text{EPI} \end{aligned}$$

②



5-P.I

$$\Sigma m(5, 7, 13, 15) \rightarrow \text{EPI}$$

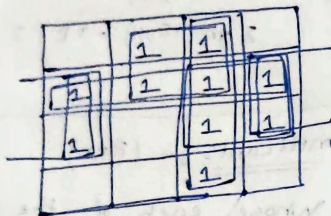
$$\Sigma m(5, 1) \rightarrow \text{EPI}$$

$$\Sigma m(13, 12) \rightarrow \text{EPI}$$

$$\Sigma m(17, 8) \rightarrow \text{EPI}$$

$$\Sigma m(8, 10) \rightarrow \text{EPI}$$

③



5-P.I

$$\Sigma m(1, 3, 5, 7) \rightarrow \text{EPI}$$

$$\Sigma m(4, 5, 6, 7) \rightarrow \text{RPI}$$

$$\Sigma m(3, 7, 14, 15) \rightarrow \text{EPI}$$

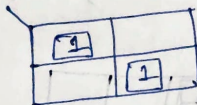
$$\Sigma m(6, 7, 14, 15) \rightarrow \text{RPI}$$

$$\Sigma m(4, 12, 6, 14) \rightarrow \text{EPI}$$

~~$\Sigma m(1, 3, 5, 7)$~~
 ~~$\Sigma m(4, 5, 6, 7)$~~

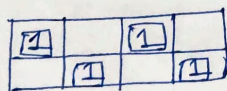
(*) Maximum No. of EPI :-

$\Rightarrow$ 2-Variables :-



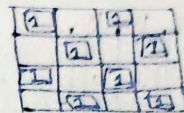
$\rightarrow 2 \text{ EPI}$

3-Variables :-



$\rightarrow 4 \text{ EPI}$

Variables :-

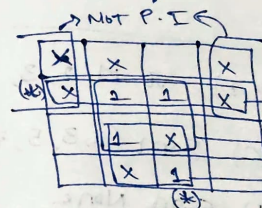


$$n \text{ Variables} \Rightarrow 2^n / 2 \text{ EPI} \approx 2^{n-1} \text{ EPI}$$

Some example with don't care (X) :-

- $\rightarrow$ Alone don't care will not form group.
- $\rightarrow$ use don't care in clubbing if it helps in increasing the group size.

Eg:-



3 P.I

2 EPI

1 RPI

Assignment-2

(a) The minimized expression for given k-map is

AB	00	01	11	10
00	0	0	1	1
01	0	X	1	X
11	X	X	X	1
10	1	0	1	1

(a) $A + \bar{B}C$

(b) $B + Ac$

(c) $C + AB$

(d) ABC

A.

AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$\rightarrow A + \overline{B}C$ [a].

(Q)

Minimized exp?

A.

AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$\Rightarrow C + \overline{B}D$

(Q) Complement of $f(x,y,z) = \pi M(2,3,4,6)$ is

(a) $\sum m(2,3,4,6)$

(c) $\sum m(1,3,5,7)$

(b) $\sum m(1,5,7)$

(d) ~~None~~ None

A.

$f(x,y,z) = \sum m(0,1,5,7)$

$= \sum m(2,3,4,6)$ [a].

(Q) $f(A,B,C) = \sum m(1,2,3,5,6,7)$ is equal to?

BC	00	01	11	10
A	1	1	1	1
A	1	1	1	1

$y = A + C + \overline{B}$

Find minimum number of 2 i/p NOR gate

required to implement following function.

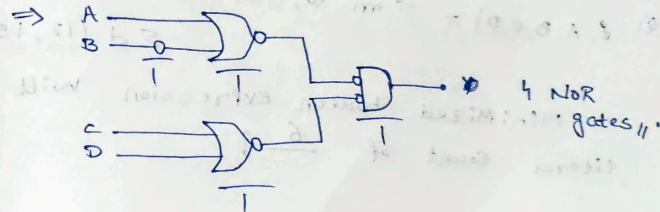
$f = ABC + AC + \overline{B}C$

NOR gate $\Rightarrow$ POS.

BC	00	01	11	10
A	0	1	0	0
A	0	1	1	1

BC	00	01	11	10
A	0	1	1	1
A	0	1	1	1

$y = (B+C) \cdot (A+\overline{B})$



4 NOR gates

(*) I, P.I, E.P.I, R.P.I, S.P.I are defined only for min terms.

(a) Find No. of EPI.

AB \ CD	00	01	11	10
00			0	
01	0	0	0	
11		0	0	0
10		0	0	

(a) 4 (b) 5 (c) 6 (d) 3

AB \ CD	00	01	11	10
00				1
01	1	1		
11				1
10				1

4 - P.I

4 - E.P.I

(Q) $f(A, B, C, D) = \sum m(0, 1, 3, 4, 5, 8, 9, 11) + \sum d(12, 15)$

Minimized boolean expression will have literal count of 6.

AB \ CD	00	01	11	10
00	1	1	1	
01	1	1		
11	X		X	
10	1	1	1	

$$Y = \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{B}D \quad L.C = \underline{6}$$

(*) If Don't cares are present, SOP & POS give you diff. Boolean expressions.

$$F(A, B, C, D) = \pi m(2, 6, 7, 10, 13, 14) \cdot \pi d(12, 15)$$

AB \ CD	00	01	11	10
00				0
01			0	0
11	X	0	X	0
10				0

$$Y = (\bar{A} + \bar{B}) \cdot (\bar{B} + \bar{C}) \cdot (\bar{C} + D) \rightarrow L.C = \underline{6}$$

(*) Prefer L.C. which is less.

(a) you are having 4 Variables

(a) Max. possible boolean exp. is 2^4 .

(b) " " Self dual fn. " y .

(c) " " P.I. is 2^4 .

(d) " " E.P.I " W .

$$x+y+z+W=8$$

A. 4-variables

$$(a) 2^4 = 2^4 = 2^{16} = 2^6 \times 2^{10} = 64 \times 1024 = 65536$$

$$(b) 2^{2^{n-1}} = 2^{2^3} = 2^8 = 256$$

(c) Prime implicants for 4-variables = 13

$$(d) E.P.I = \frac{2^4}{2} = \frac{16}{2} = 8$$

$$64 + 256 + 13 + 8 = 341$$

$$\Rightarrow 64 + 256 + 13 + 8 = 341$$

$$f = f_1 \cdot f_2 \quad (mca)$$

$$f_1 = \sum m(0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 15)$$

$$f_2 = \sum m(4, 7, 15)$$

f_2 can be -

$$(a) f_2 = b(c+d) \quad (d) = b(\bar{c}+\bar{d})$$

$$(b) f_2 = b(\bar{c}+d)$$

$$(c) f_2 = b(c+d)$$

$$f = f_1 \cdot f_2$$

$$\Rightarrow f_2 = \sum m(\dots, 4, 7, 15)$$

(Allowable Set) 5, 6, 12, 13

$$(a) f_2 = bc + b\bar{d}$$

	<u>bc</u>				<u>b\bar{d}</u>			
	00	01	11	10	(6) 0110	(4) 0100	(5) 0110	(1) 0100
A					(14) 1110	(10) 1100	(12) 1100	(11) 1100
0					(7) 0111	(13) 1111	(15) 1111	(3) 0111
1								

$$(b) f_2 = b\bar{c} + bd$$

	<u>b\bar{c}</u>				<u>bd</u>			
	01	11	00	10	0101 (5)	1101 (13)	0111 (7)	1111 (15)
					0100 (4)	1100 (12)	0101 (5)	1100 (12)
					0110 (6)	1110 (14)	0110 (6)	1110 (14)

[a, b].

Check those numbers are present in allowable f_2 set (or) not.

(a) The boolean expression, $f = \overline{AB}BC + CA$ is equal to

- (1) $\overline{AB} + \overline{B}C + \overline{C}A$ (b) $\overline{AB} + \overline{B}C + \overline{A}B\overline{C}$
 (c) $(\overline{A} + \overline{B})(\overline{C} + \overline{B} \cdot \overline{A})$ (d) $\overline{AB + C(A \oplus B)}$

A.

$$f = \overline{AB}BC + CA$$

$$f^D = f \Rightarrow f' = \overline{A} \cdot \overline{B} + \overline{B} \cdot \overline{C} + \overline{C} \cdot \overline{A}$$

[a]

$$(b) \overline{AB} + \overline{B}C + \overline{A}B\overline{C}$$

$$= \overline{A} [\overline{B} + B\overline{C}] + \overline{B}C$$

$$= \overline{A} [\overline{B} + \overline{C}] + \overline{B}C$$

$$(c) (\overline{A} + \overline{B})(\overline{C} + \overline{B} \cdot \overline{A})$$

$$= \overline{A}\overline{C} + \overline{B}\overline{C} + \overline{A}\overline{B} + \overline{A}\overline{B}$$

$$= \overline{AB} + \overline{B}C + \overline{C}A$$

$$(d) g = \overline{AB} + C(A \oplus B)$$

$$= \overline{AB} + C\overline{A}B + CAB$$

$$\Sigma m(6, 7, 3, 5)$$

Check Minterms of Both functions.

$$f = \Sigma m(6, 7, 3, 5)$$

END OF
DAY-12